

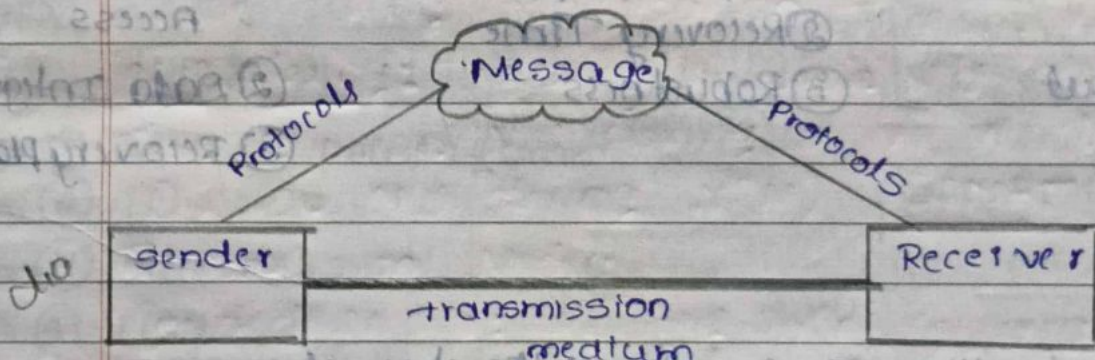
DATA COMMUNICATION

2M Data communication definition

Data communication is the exchange of digital information between devices via transmission media using standardized protocols (rules & regulations in communication)

IMP 5M 12M Components of Data Communication

- ① Message :- Data/Information being transmitted
- ② Sender : Device that originates transmission
- ③ Receiver : Device that receives transmission
- ④ Transmission Medium : Physical path (cable, wireless)
- ⑤ Protocols : Rules governing communication



IMP 5M Type of data communication / logical transmission

Transmission mode

Simplex

only one direction

eg:- TV broadcast.

Duplex

both direction

→ **Half Duplex**

both directions but in 1 direction at a time

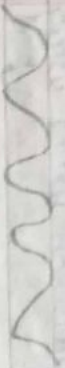
→ **Full Duplex**

both direction at the same time

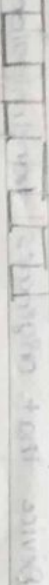
eg:- telephone, modern networks.

Transmission Methods

① Analog signal → continuous signals
(voice, traditional radio)



② Digital signal → discrete binary signals
(computers, modern networks)



Key Criteria for Transmission Methods

Criteria

Performance	Reliability	Security
-------------	-------------	----------

- ① Bandwidth: the amount of data that can be sent over a link in a given time
- ② Latency: the time it takes for data to travel from source to destination
- ③ Throughput: the amount of data that is actually sent over a link in a given time
- ④ Factors: the factors that affect the performance of a transmission method

Performance

- ① Bandwidth: the capacity of the channel
- ② Latency: the data must arrive in an acceptable amount of time
- ③ Throughput: the actual speed and accuracy at which data travels from source to destination
- ④ Factors: number of users sharing the same medium, hardware capabilities (speed of switches/routers etc), network topology

14.1.2.1

Network Topology

def a network is the arrangement of devices (nodes) and connections (links) in a computer network. It shows how computers, servers and other devices are connected and how data flows between them.

Types of Network Topology

① Physical Topology

Refers to the actual physical layout of the network, regardless of how the data is actually transmitted. It shows the physical arrangement of the network, including the location of the devices and the physical connections between them.

② Logical Topology

Refers to the way the data is actually transmitted, regardless of the physical layout of the network. It shows the logical arrangement of the network, including the logical connections between the devices.

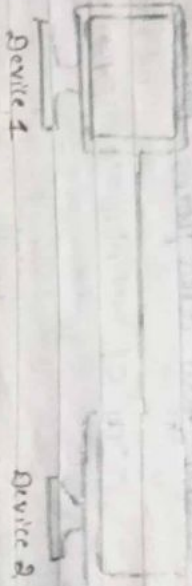
Types of Network Topology

- ① Point-to-point (peer-to-peer)
- ② Bus
- ③ Star
- ④ Ring
- ⑤ Mesh
- ⑥ Hybrid
- ⑦ Tree

Network topology is the arrangement of computers and devices in a network which shows how they are interconnected and how data flows between them. It shows the physical and logical arrangement of the network and how the devices are connected to each other.

① Point-to-Point (Peer-to-Peer) Topology

Key characteristics and Features: Type of topology in which 2 devices are directly connected with a single cable.



- Redundant connection: The link is used exclusively by the two nodes with no interference from other devices.
- Simple structure: Direct physical or logical connection (e.g., serial cable, fiber optic)
- Performance: Offers high, consistent bandwidth and low latency
- Scope: Typically used for short-distance, direct connections

② Advantages:

- ① High speed / Performance: Since the bandwidth is not shared, data transmission is fast and efficient
- ② Easy Setup / Maintenance: Simple to configure and troubleshoot

- ③ High security / Reliability: Private connection minimizes interception risks and provides high reliability

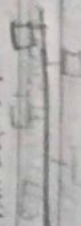
Disadvantages:

- ① Limited scalability: Not suitable for connecting many devices; difficult to expand
- ② Single Point of Failure: If the connection link fails, communication between the 2 nodes stops entirely

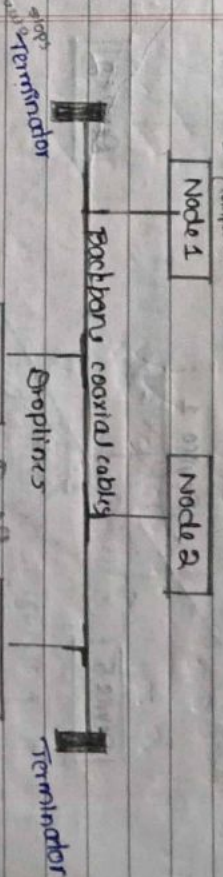
- ③ High cost per node: Expensive to set up for many devices compared to other topologies.

② Bus Topology

Key characteristics and components



- ① Single Backbone: All nodes share a common communication medium. [Data type, all topology types are divided into a single main cable]
- ② Broadcast system: When a node sends a message, it travels in both directions and is received by all other stations but only the intended recipient accepts it
- ③ Terminators: Required at both ends of the main cable to absorb signals and prevent them from bouncing back
- ④ Linear structure: Devices are connected in a line, often using coaxial cable



Advantages

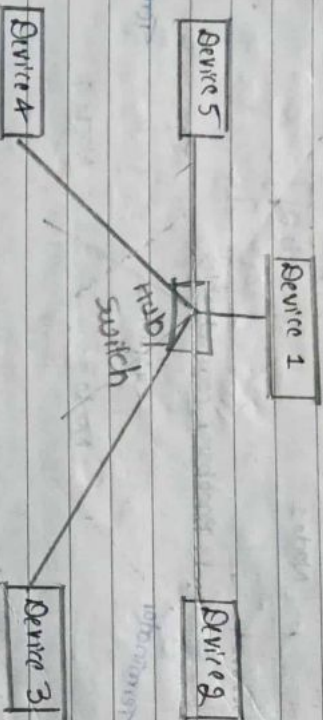
- ① Low cost: Regular minimal cabling compared to other topologies, such as star networks
- ② Simplicity: Easy to set up, extend and manage for small networks
- ③ No central Hub: No hub requires a central hub or switch for connectivity

Disadvantage

1. Single point of failure: In the main cable (backbone) fails, the entire network goes down.
2. Limited scalability: Not ideal for large networks; performance decreases significantly with more devices.
3. Troubleshooting difficulties: Identifying faults in specific nodes or the cable can be difficult.
4. Security & efficiency: Since all nodes receive all traffic, it is less secure and prone to data collisions.

- ③ Star Topology:- In this type of topology, all devices are connected to a single central device through a cable.
- This hub is the central node and all other nodes are connected to the central node.

Star Topology



- ### Advantages
- Easy to install and modify.
 - Failure of one cable does not affect others.
 - Centralized monitoring and management.
 - Easy to troubleshoot.

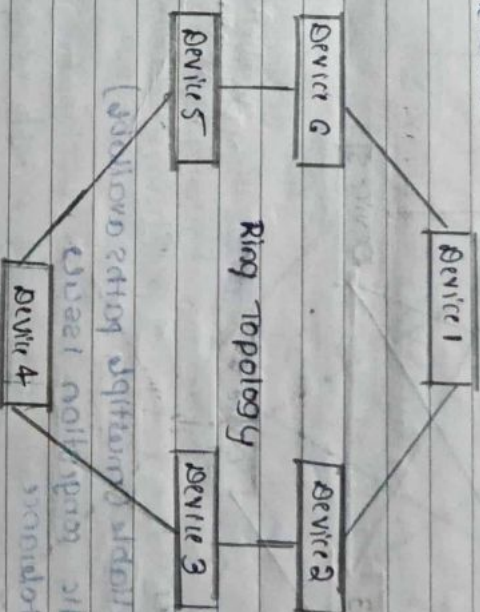
Disadvantage

- Central hub is single point of failure.
- Requires more cable than bus topology.
- More expensive due to hub/switch.

- ④ Ring Topology - In this type of topology, all devices are connected in a single loop, forming a ring connecting devices with exactly two neighbouring devices.

- A number of repeaters are used for Ring topology with a large number of nodes, because if someone wants to send some data to the last node in the ring, topology with 100 nodes, then the data will have to pass through 99 nodes to reach the 100th node.
- Hence to prevent data loss, repeaters are used in the network.

Ring Topology



Advantages

- Ordered network access prevents collisions.
- Equal access for all devices.
- Performance does not degrade with heavy traffic.

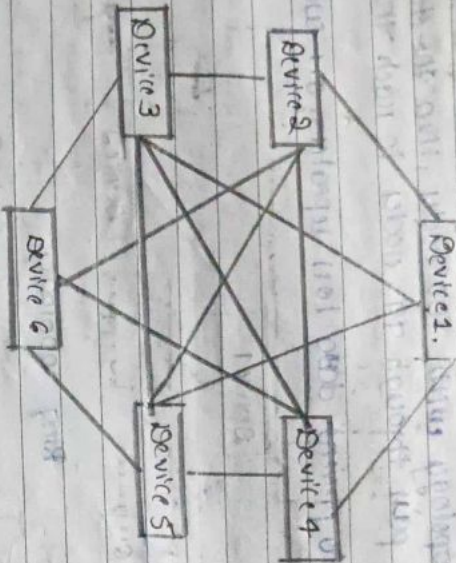
Disadvantage

- Failure of one node can break entire network.
- Difficult to troubleshoot.
- Adding/removing devices disrupts network.

⑤ Mesh Topology

A network configuration where every device is interconnected with every other device.

- The nodes are connected to each other completely via a dedicated ~~to~~ link during which information travels from nodes to nodes
- Each computer not only sends it signals but also transfer data from other computers
- The connections within the mesh are often wired or wireless



Advantage

- Highly reliable (multiple paths available)
- No traffic congestion issues
- Fault tolerance

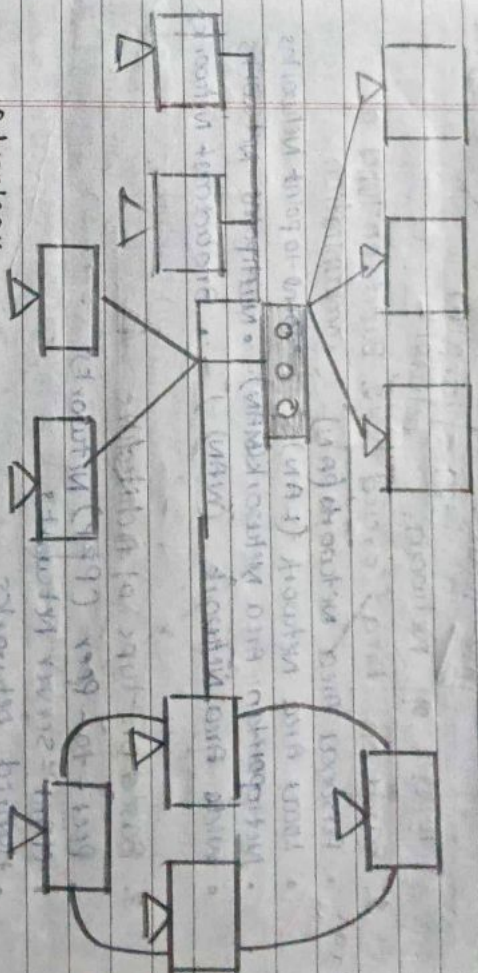
Disadvantage

- Expensive (lots of cabling and ports) required
- Difficult to install and manage and loops
- Complex configuration for large networks

⑥ Hybrid Topology

A network topology that combines two or more different network topologies.

- A combination of bus topology, ring topology and mesh topology.
- The structure of hybrid topology is more complex but offers various advantages such as flexibility and fault tolerance.



Advantage

- Flexible and scalable.
- Can be optimized for specific needs.
- Reliable (Benefits of combined topologies).

Disadvantage

- Complex design & implementation
- Expensive & slower compared to other topologies
- Difficult to manage (too various how system)

⑦ Tree Topology

Tree topology is a type of topology which has a structure like a tree with root nodes & branches.

Network Type

A computer network is a group of two or more independent computers and devices connected to share data, resources and services. These connections can be established using wired media or wireless technology.

The criteria of distinguishing one type of network from another is different and sometimes confusing.

Q1. What is computer network? Explain different types of network.

- Types of network:
1. Based on Area (covered)
 2. Based on type of communication

- Personal Area Network (PAN)
- Local Area Network (LAN)
- Metropolitan Area Network (MAN)
- Wide Area Network (WAN)
- Broadcast Networks
- Peer-to-Peer (P2P) Networks (1 to 1)
- Client-Server Networks (requiring software to manage)
- Hybrid networks

[Both are covered by application of]

1. Based on Area Covered

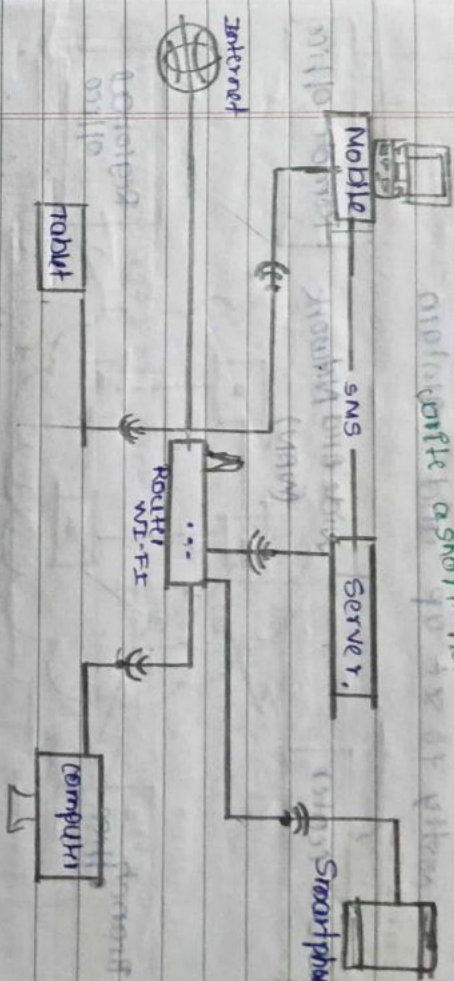
1. Personal Area Network (PAN)

A PAN is a short-range network that connects personal devices like smartphones, tablets and computers. It typically covers less than 10 meters and usually uses wireless technology such as Bluetooth.



2. Local Area Network (LAN)

- A LAN connects computers and devices within a small area like a home, office, school or hospital.
- It usually uses switches, routers and private IP addresses.
- LANs are high speed, inexpensive (cost up to 100 Mbps) and maintain communication up to 1 km.



3. Metropolitan Area Network (MAN)

- A MAN spans 5-50 km, covering more area than a LAN but less than WAN.
- It connects computers across a city or between nearby cities.
- MANs provide high speed connectivity (in Mbps).
- Can act as ISPs and are useful for organisation needing fast communication.



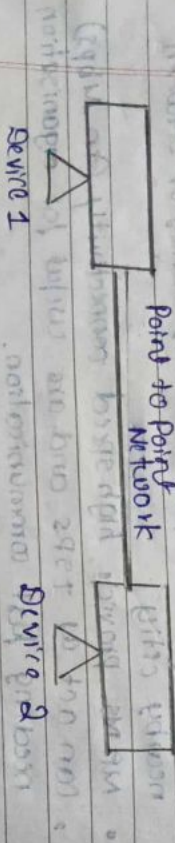
4. Wide Area Network (WAN)

- A WAN covers large geographical areas (hundreds of km)
- often connecting multiple LANs through telephone lines, radio waves or satellite
- It can be private (for organisations) or public (like the Internet)
- WANs offer high-speed communication but are costly to set up and maintain

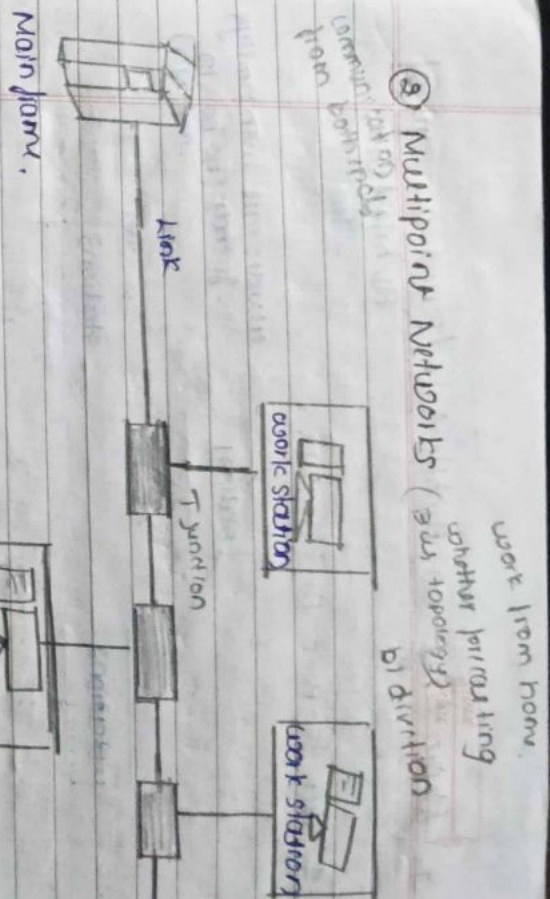


2. Based on type of communication

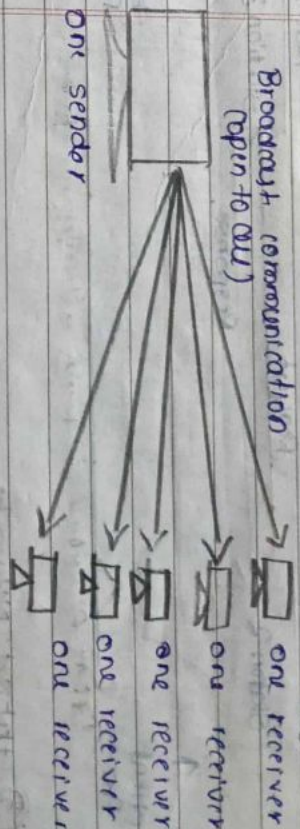
- ① Point to point Networks • sending msg. in some of point to point topology in a direction



② Multipoint Networks (bus topology) b) division

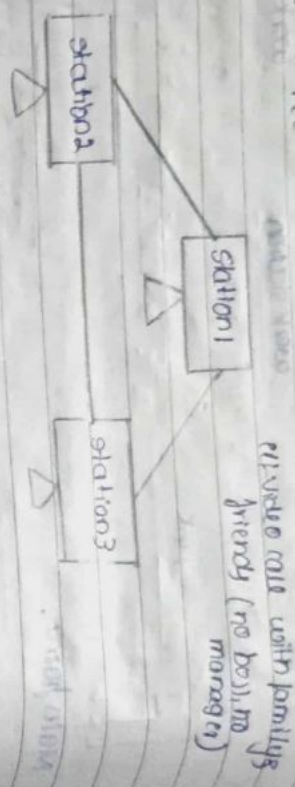


③ Broadcast Network one sender, multiple receiver

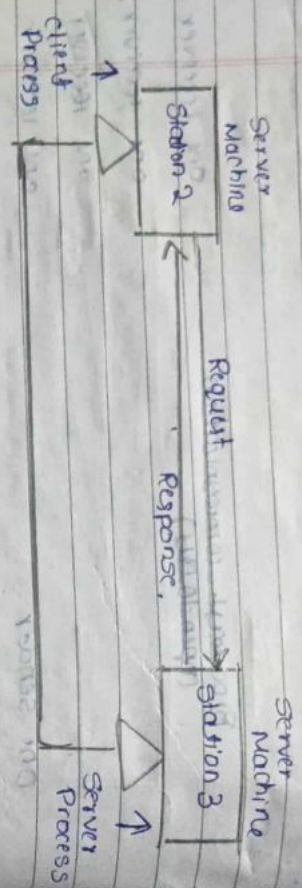


3. Depending on Architecture / Based on Type of Architecture.

1. Peer-to-Peer Architecture.

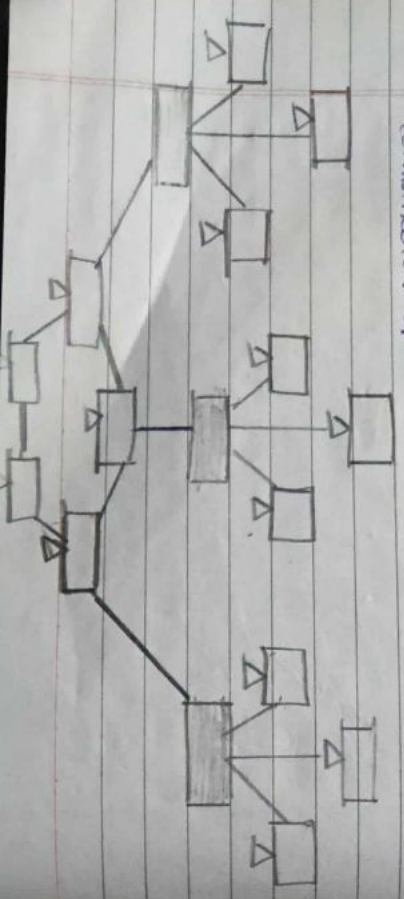


2. Client / Server Architecture. on system browser like client



3. Hybrid Network

combination of all type of network



The Internet

Internet is a global communication system that links together thousands of individual networks.

- It allows exchange of information plus 2 or more computers on a network. Thus Internet helps in transfer of messages through mail, chat, video & audio conference, etc.
- It has become mandatory for day-to-day activities: bills payments, online shopping and surfing, tutoring, working, communicating with peers etc.

explain ARPANET & CSMA with an example on ARPANET-SEA

History of Internet

Internet was evolved in 1969, under the project called **ARPANET** (Advanced Research Projects Agency Network) to connect computers at different universities and US defense.

- Soon after the people from different backgrounds such as engineers, scientists, students and researchers started using the network for exchanging information and message.

In 1990s the Internet working of ARPANET, NSFnet and other private networks resorted into **Internet**.

Therefore, Internet is a global network of computer networks.

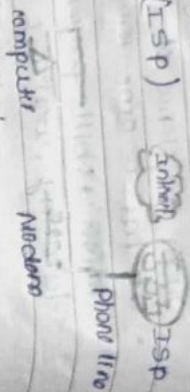
Desktop computers, mainframe GPS... connected to net.

net could operate through the world's...

off computer has longer life expectancy...

How do we connect to the Internet?

- Computer
- Connection - phone line, cable, DSL, wireless...
- Modem
- Network software - TCP/IP
- Application software - web browser, email, ...
- Internet Service Provider (ISP)



What can we do on the Internet?

- Send and receive email messages
- Download free software with FTP (File Transfer Protocol)
- Post our opinion to a Usenet newsgroup
- Surf the world wide web
- And much, much more...
- There is no charge for most services

How does the Internet work?

The Internet operates using a client-server model:

Client: A device such as a laptop or smartphone connected to the Internet

Server: Powerful computers that store websites and data.

Step by step working

1. The client sends a request through a browser
2. The server identifies the websites using its IP address
3. DNS converts the domain name into an IP address
4. The request is routed to the correct server
5. The server processes the request and returns the webpage.

What is IP address? with diagram explain different kinds of IP address.

What is an IP Address?

An IP Address (Internet Protocol Address) is a unique identifier assigned to every device connected to the Internet by an ISP (Internet Service Provider).

- It works like a digital identity, similar to Aadhaar or PAN

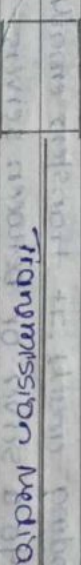
Example format: 192.154.3.89

Range: 0.0.0.0 to 255.255.255.255

Network Protocol Explain protocols & standards.

Network protocols comprise mechanisms for devices to recognise and make connections with each other and as formatting rules, which specify how data is packed into messages sent and received.

For communication using secure networking such as networks protocols cannot be ignored.



Sender

Receiver

Network protocols are the rules that help devices connect & communicate, recognize each other, in a network.

Protocol	Others
① TCP	① SMTP
② IP	② POP3
③ HTTP	③ DNS
④ HTTPS	④ SSH
⑤ FTP	⑤ SSH

Basic + protocols in Networking

Important

Protocols used in Computer Networking

I. Communication Protocol

① TCP (Transmission Control Protocol)

Ensures data is sent and received accurately by breaking it into packets, sending them, and reassembling them at the destination.

② IP (Internet Protocol) provides

addresses and routes the packets to make sure they reach the right destination.

II. Web Protocol

③ HTTP (Hypertext Transfer Protocol/Service)

HTTP used for transferring webpages on the internet.
 HTTP → open website
 internet → open website instantly

III. File Transfer Protocol

④ FTP (File Transfer Protocol)

used for transferring files between computers on a network.
 it allows user to upload & download files.

IV. Other Email Protocol

⑤ SMTP (Simple Mail Transfer Protocol) email related

used for sending emails. It transfers emails from a client to a server or between servers.

⑥ POP3

⑦ DNS (Domain Name System)

It is used to translate human-readable domain names (like www.example.com) into IP addresses that computers use to identify each other on the network.
 human-readable domain name → IP address
 translate website name into IP address

⑧ DHCP (Dynamic Host Configuration Protocol)

Automatically assigns IP addresses to devices on a network, ensuring each device has a unique IP address.

⑥ POP3 → used for retrieving email

provides a secure way to access and manage devices over a network. It encrypts the data, making it safe from eavesdropping.

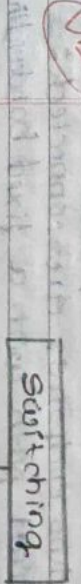
SSH (Secure Shell) → secure login to another computer.

⑤ SNMP (Simple Network Management Protocol)

used for managing and monitoring network devices like routers, switches & servers.

Switching

an internet is a switched network in which a switch connects at least 2 links together.
 A switch needs to forward data from a network to another network action required.



Circuit

Packet

Message

Packet Switching

store & forward technology. Signal network that transfers data files by breaking them up into smaller units ("packets") and transmitting them through a series of nodes ("switches").

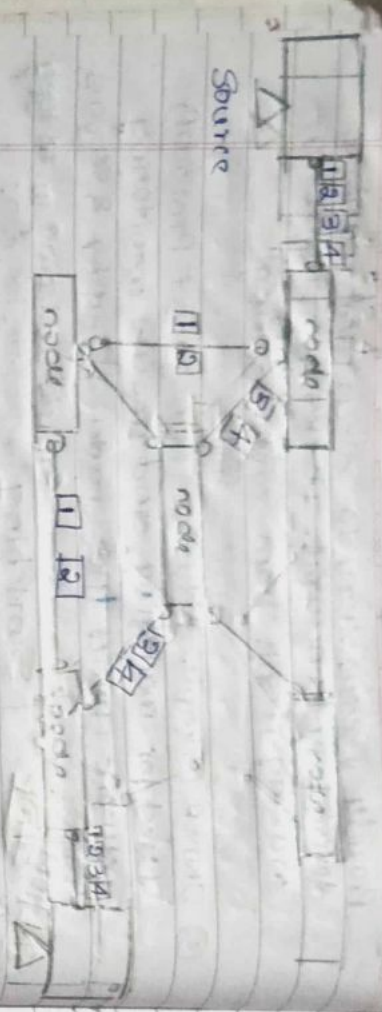
In a packet known as store-and-forward the switches route the packets to their destination where the data files are then reassembled.

The internet and most local area networks (LANs) are packet-switched networks.

At packet level called store-and-forward.

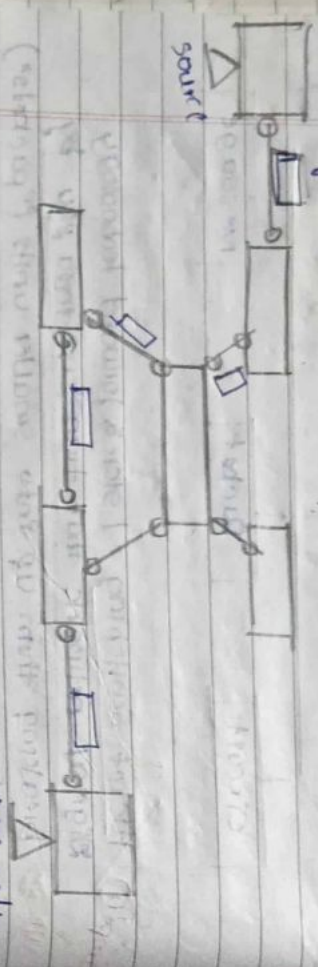
Packets are sent one by one to the destination. At end all packets are joined to form original data.

Message switching with store and forward



Circuit Switching

Technology used for landline telephony and data networks where two endpoints are connected (dedicated) by a single physical path with a fixed bandwidth.



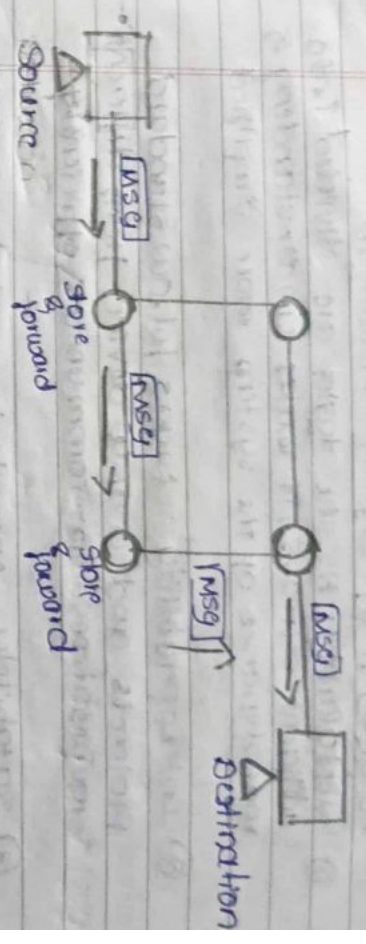
Message Switching

Message switching is a connectionless technique in which the entire message is routed from the source node to the destination node by one hop at a time, called hop-by-hop system.

Message switching is a digital network in which the entire message is sent at once.

Each message is stored and then forwarded.

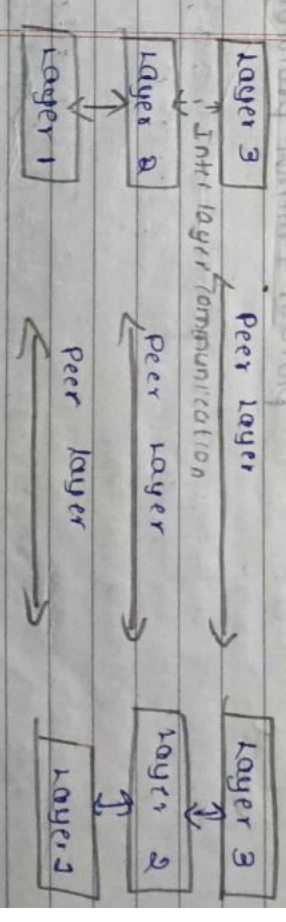
③ No valid path, message is dropped.
④ It also uses store and forward method.
⑤ Take more time compared to packet switching.



Network Models

Layered Architecture

- Each layer assigned a specific set of tasks and responsibilities.
- Ensures modularity, flexibility and easier troubleshooting.
- Each layer operates independently and prepares for assigned functions while depending on adjacent layers only for inputs and outputs.



Advantage of layered architecture

- ① Modularity: As the tasks are divided into different sections, it makes understanding & maintenance of the system more simplified
- ② Interoperability: Layers follow standard protocols and enable devices from different organisations to communicate efficiently
- ③ Scalability - New technological protocols can be integrated without affecting the entire system
- ④ Troubleshooting - Problems can be isolated to specific layers & each layer can be analyzed and tested individually

The most commonly used Architecture are:
 ① OSI model (Open systems Interconnection)
 ② TCP/IP model (Transmission control protocol Internet protocol)

up to 7 layers
 7 layers
 OSI Model (Open system Interconnection)
 TOP most

Application layer

Responsible for providing services to the user

Presentation layer

Take care of syntax and semantics of the information exchange between 2 communication system

Session layer

→ It establish, maintain, synchronize and terminate the interaction between sender and the receiver

Transport layer

→ Responsible for providing reliable delivery of data (packets)

Network layer

→ Responsible for delivery of individual packet from source to destination

Data link layer

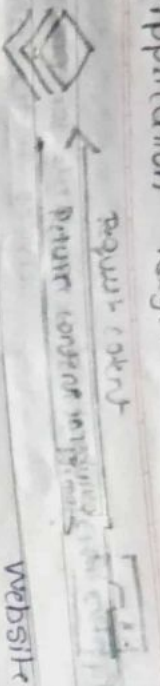
→ Responsible for moving frame from one hop to next hop

Physical layer

→ Responsible for moving individual bits from one device to the next device.

Explain OSI reference model in detail.

7. Application Layer

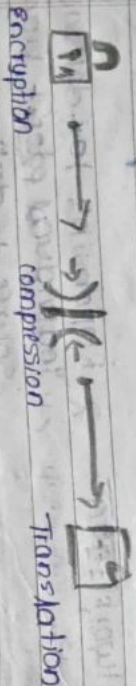


Application Layer

- This is where the user interacts with the data through apps like browsers, email, or games.
- Examples of protocols operating at the Application layer include HTTP, HTTPS, FTP.

- ① Application layer for web browsing
- ② File Transfer Protocol (FTP) for file transfer
- ③ SMTP - Simple Mail Transfer Protocol for email services
- ④ DNS (Domain Name System) for resolving domain name to IP address

6. Presentation Layer



3 main tasks

- ① Data Translation: converts data between different formats to handle compatibility between heterogeneous system
- ② Data Encryption & Decryption: Secure data by converting it into an unreadable format during transmission and reversing it back at the destination
- ③ Data Compression: Reduce the size of data to save bandwidth and improve transmission speed.

5. Session Layer

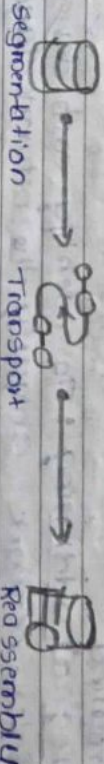


session of communication

- Manages and controls the connections between computers
- It establishes, maintains, and terminates connections ensuring that data exchange occurs efficiently and in organized manner.
- Dialogue Control: It determines how systems communicate, managing turns through

- ① Simplex: One-way communication
- ② Half-Duplex: Two-way, but only one side transmits at a time
- ③ Full Duplex: Both sides transmit simultaneously

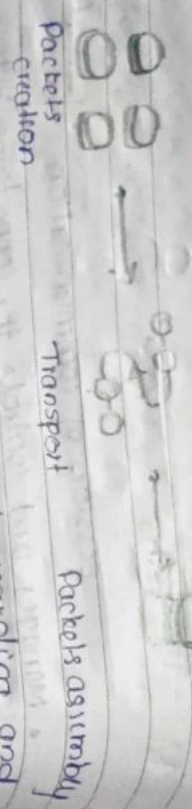
4. Transport Layer



explain transport layer protocols

- Provides end-to-end communication services for applications
- It ensures complete data transfer, error recovery and flow control between hosts.
- This layer segments and reassembles data for efficient transmission & provides reliability via error detection and correction mechanisms.
- Connection Control: It manages 2 types of data flows
- ① Connection-oriented: - Requiring a 'handshake' (e.g. TCP) - to ensure reliable delivery. eg. web browsing and email.
- ② Connectionless: - Sends data without prior connection (e.g. UDP User Datagram Protocol) prioritizing speed over reliability (e.g. - video streaming & online gaming)

3. Network Layer.

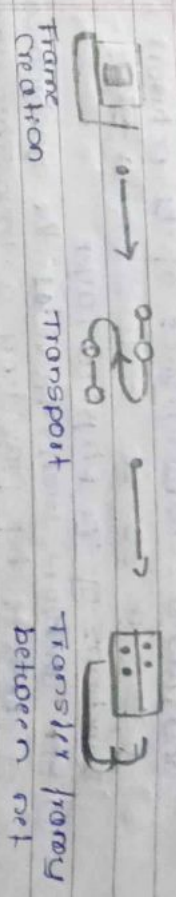


- Responsible for data routing, forwarding and addressing
- It determines the best prepared path for data to reach its destination based on network conditions the priority of service and other factors
- This layer manages logical addressing through IP addressing and handles packet forwarding

Functions of Network Layer

- Routing - The network layer protocols determine which route is suitable from source to destination. This function of the network layer is known as routing
- Logical addressing - To identify each device in the network uniquely, the network layer defines an addressing scheme.
- The sender and the receiver's IP addresses are placed in the header by the network layer
- Such an address distinguishes each device uniquely and universally

2. Data Link Layer.

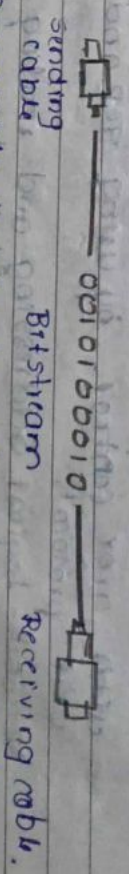


- Responsible for node-to-node data transfer and error detection and correction.
- It ensures that data is transmitted to the correct device on a local network segment.
- This layer manages MAC media Access control address and is divided into two sub-layers:

- 1) Logical Link Control (LLC) and
- 2) Media Access Control (MAC)

- When a packet arrives in a network, it is the responsibility of the LLC to transport it to the host using MAC address
- Packet in the Data Link layer is referred to as frame
- The packet received from the Network Layer is further divided into frames depending on the frame size of the NIC (Network Interface Card)
- LLC also encapsulates sender and receiver's MAC address in the header

1. Physical Layer



- Responsible for the physical connection between devices
- It defines the hardware elements involved in the network, including cables, switches and other physical components
- This layer also specifies the electrical, optical

and radio characteristics of the network

Functions of the Physical Layer

- ① BPS Synchronization: Provides the synchronization of the bits by providing a clock. This clock controls both sender and receiver thus providing synchronization at the bit level.
- ② BPS Rate control: Defines the transmission rate, i.e. the number of bits per second.
- ③ Physical Topology: Specifically how the duplexing, device modes are arranged in a network, i.e. bus topology, star topology or mesh topology.
- ④ Transmission mode: Also defining how the data flows b/w the 2 connected devices. The various transmission modes possible are simplex, half-duplex & full duplex.

② TCP/IP

Application: presents data to the users, encoding and session control data translation with the help of protocols, i.e. HTTP, FTP, DNS, DHCP.

Transport: support used to end connection establish ment of data segments and delivery with error control, by using TCP and UDP protocol.

Network: logical addressing and routing of data packet from source to destination by identifying neighbours
Data link: combine raw data into frames, error recovery and retransmissions

Physical: provide the physical interface for data transmission.

Chapter 4
Medium Access Control Protocol (MAC)
(multiplexing) → Take parts, group, multiplex

OSI Reference Model. (This diagram is compulsory)

Application Layer	Layer 7
Presentation	Layer 6
Session	Layer 5
Transport	Layer 4
Network	Layer 3
Data Link	Layer 2
Physical	Layer 1

↑
compulsory 7 layers in OSI reference model in computer network
Line coding solution

① Explain CSMA/CD protocol in detail. (CSMA/CD is a MAC protocol used in detail. CSMA/CD is a MAC protocol used in detail.)

② TCP/IP reference model. (TCP/IP reference model is a 4-layer model. It is a reference model for the Internet.)

③ Explain SDN architecture. (SDN is a network architecture for software-defined networking.)

④ Explain Aloha protocol in detail. (Aloha is a MAC protocol used in detail. It is a simple protocol for CSMA.)

⑤ What is CSMA and explain types of CSMA in detail. (CSMA is a MAC protocol used in detail. It is a simple protocol for CSMA.)

⑥ Explain following: (a) shortest path routing (b) optimal routing

⑦ What are the causes of congestion? Explain any one algorithm for controlling congestion.

⑧ Write a short note on UDP, IP header, and B-TSUN, CSMA, cellular system, CRU.

⑨ Explain transport layer protocol.

⑩ Application of computer network.

DATA COMMUNICATION

classmate

Date _____
Page _____

2M

Data Communication: Data communication is a process of exchanging data between two or more devices through a wired or wireless transmission medium using standard protocols.

Explain data communication components with diagram

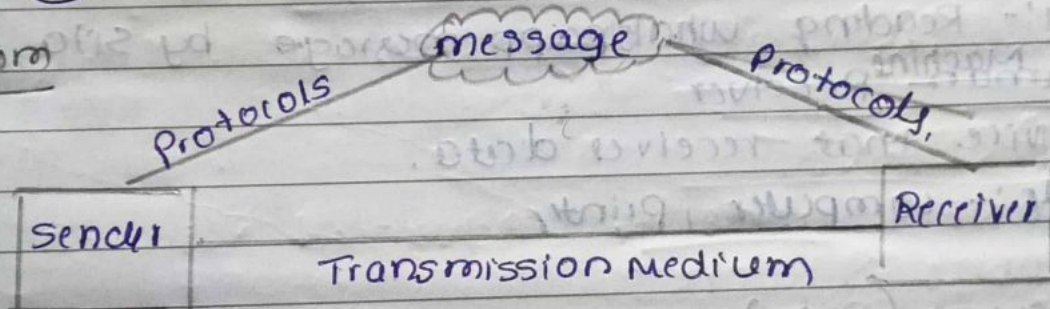
10M

Components of Data Communication

There are 5 main components

- ① Sender (source)
- ② Receiver (destination)
- ③ Message (Data)
- ④ Transmission medium (channel) / communication channel
- ⑤ Protocol

Diagram



- ① **Sender**
 - Sender is a device that sends the data
 - It starts communication process
 - It creates message known as "source of data"
 - Sender can be computer/mobile.
 - Converts data into signals
 - Sends data through medium
 - Data can be text, audio, video or images

Types of Sender

① Human Sender :-

• The person who sends data
ex:- typing message sending email.

② Machine Sender

• Device that send data automatically
ex:- computer, sensor, mobile app.

② Receiver (Destination)

1. Receiver is the device that receives data
 2. It is end point of communication. Ends communication
 3. It receives/accepts data from sender
 4. It converts signal to readable data
 5. It can be computer, mobile, printer
 6. It may store or display data
- ex:- mobile receiving whatsapp message

Types

① Human Receiver

- Person who receives and reads data
- are called human receiver

ex:- Reading whatsapp message by sita

② Machine Receiver

- Device that receives data.
- ex:- computer, printer

③ Message (Data)

1. Message is the information being sent
2. It is main content of communication
3. It can be text, image, audio or video
4. It is also called as Data
5. It is converted into signals for transmission
6. It is generated by sender
7. It travels through medium
8. It may be encrypted for security

ex:- email, file

Types of Message

① Text

- Written information
- ex:- SMS, email

② Audio

- sound data
- ex:- voice call

③ Video

moving image + sound

ex:- youtube video

④ Image

picture, data

ex:- photos

⑤ Transmission Medium (Channel)

1. It is a physical path for data transfer
 2. Connect sender and receiver
 3. It can be wired or wireless
 4. Carry signals from one device to another
 5. Affects speed of communication
 6. Affects data quality
 7. It can be guided or unguided medium
 8. It can be wired or wireless
 9. Wired example: cable, fiber optic
 10. Wireless: wifi, radio waves
- ex:- Internet network

Types of Transmission medium

① Guided (wired) →

② Twisted pair cable

③ Coaxial cable

④ Fiber optic cable

⑤ Infrared

⑥ Microwave

⑥ Protocol

Network protocols are the rules that help devices communicate with each other.

1. Protocol is a set of rules for communication
2. Controls how data is sent & received
3. Avoids error in communication
4. Defining format of data
5. Provides error checking
6. Essential for network communication

Types of Protocols

- ① Communication protocols
- ② File Transfer Protocol
- ③ FTP (File Transfer Protocol)
- ④ Email protocols
- ⑤ Web protocols
- ⑥ SMTP (Simple Mail Transfer Protocol)
- ⑦ POP3 (Post Office Protocol version 3)
- ⑧ HTTP
- ⑨ HTTPS

Types of LAN

Transmission Media

Network Topology

Network types

ARPANET

① ARPANET stands for Advanced Research Project Agency Network

② It was first packet-switching computer network developed in the late 1960s by the US Department of Defense (ARPA)

③ It is considered as the foundation of modern Internet.

④ ARPANET (Advanced Research Project Agency Network) is the first computer network developed by the US Department of Defense that connect computers & it is the foundation of network

purpose

① Developed in 1969 by US Department of Defense

1. First packet-switching network
2. Connected only research institutions initially
3. Used TCP/IP protocol
4. Allowed communication b/w different computer system.

5. Introduced concept of 'decentralized networking'
6. Data was sent in small packets instead of full message
7. Packets travel through different paths.
8. Receiver rearrange them into original message.

(Simple and this) ARPANET

② Full form, definition

③ Year & Origin

④ Purpose - To share information b/w research computers

• allow communication b/w distant computer

⑤ Technology used - packet switching

• Data divided into small packets before sending

⑥ Importance

• First wide area computer network

• managed even if one link fails.

• connected universities & research center

- It is the foundation of today's Internet

- introduced network concept like TCP/IP.

⑦ First connection - UCLA ↔ Stanford Research Institute.

⑧ Conclusion: ARPANET is first network & the foundation of modern Internet

SM 1101

group of interconnected computer to share data, resource, information with

Application of computer Network [limited area]

defn computer network is a group of 2 or more computers connected together to share data, resource, information within limited area

Application of computer Network.

① Education

- computer network is used for online class for student so that they can attend class from home
- teachers share study materials, notes, assignments through apps
- students can attend class using Zoom, google meet, online exams and tests, quizzes can be conducted using network
- can access online library

② Banking

- ~~Bank~~ computer network helps in online banking service such as money transfer, balance check
- ATM machines are connected to network for faster withdrawal.
- customers can use mobile banking apps for transaction anytime.
- secure & fast financial transaction through network
- transaction details updated instantly through network

③ Health care.

- online appointment booking
- sharing patient reports between hospitals.
- online doctor consultation is possible.
- Over network.
- Emergency services use network for quick communication

④ Business Office

- employees communicate through email & messages in office
- Files & documents are shared easily b/w departments using network
- Business use cloud storage to access & store data securely
- video conferencing, video meeting through network

⑤ Entertainment

- users can use online game using network
- video streaming service like Youtube, Netflix through network
- music streaming apps allow user to listen songs
- social media platforms, use
- users can download movies, games

⑥ E-commerce

- users can shop online using website like Flipkart, Amazon, meesho
- computer network supports online payment apps like UPI & card.
- customer can search products & compare price online
- order tracking system help users to check delivery status
- customer reviews & rating helps customer to understand the quality about product through network

What is IP address? with diagram explain clearly
of IP address

Solution

IP address (Internet Protocol Address)

IP address is a unique number (numerical address) assigned to each device in a network to identify a computer on a network

For one device, format of IP address is written in 4 parts (octets) separated by dots '.'

eg:- 192.168.1.1

Classes of IP address (IPv4)

IP address are divided into 5 classes

- ① class A - 10.0.0.0 to 126.255.255.255
- ② class B - 128.0.0.0 to 191.255.255.255
- ③ class C - 192.0.0.0 to 223.255.255.255
- ④ class D - 224.0.0.0 to 239.255.255.255
- ⑤ class E - 240.0.0.0 to 255.255.255.255

① Class A IP addresses

Range:

Class A IP address used for very large network where first octet represents the network ID and remaining three octets represent host ID (octet means part of IP, it contains 4 parts separated by dots.)

Range : 1.0.0.0 to 126.255.255.255

• used for large networks.

• first octet (part) = Network ID

• Remaining = Host ID (remaining 3)

• Many Host, few network

Example

10.1.2.3

Here network ID is 10

host ID is 1.2.3

③ Class B IP address

Class B IP address is used for medium-sized network where first 2 octets represent networkid and remaining 2 represent the host id

Range 128.0.0.0 to 191.255.255.255

- used for medium sized network
- first 2 octet = Network ID
- Remaining 2 = Host id.

Example 172.16.5.14

Network ID = 172.16

Host ID = 5.14

③ Class C IP address

Class C IP address is used for small network where the first three octets represent the network ID and the last octet represents the host ID.

Range 192.0.0.0 to 223.255.255.255

- used for small medium network.
- first 3 octet = network ID
- last 1 octet = host ID
- Most commonly used class

Ex 192.168.1.10

Network ID = 192.168.1

host ID = 10

④ Class D IP address

Class D is used for multicasting communication where data is sent from one sender to multiple devices instead of single device

Range: 224.0.0.0 to 239.255.255.255

- used for multicasting (group communication)
- used in streaming ~~data~~ service (youtube...)
- sendy data to multiple device
- No host id in network ID.

Ex 224.0.0.1

⑤ Class E IP address

Class E IP address is reserved for experimental & research purpose and is not used in normal networking

Range 240.0.0.0 to 255.255.255.255

Ex 240.1.1.1

multicast = one sender $\rightarrow$ many receivers
should receive

Diagram for class of IP address

	1. byte 1	2. byte 2	3. byte 3	4. byte 4
class A	Network ID	Host ID		
class B	Network ID	Host ID		
class C	Network ID	Host ID		
class D	Network ID	Host ID		
class E	Network ID	Host ID		

in class A = 1st octet -> network id
 remaining 3 -> host id

B = 1st 2 octets -> net id
 remaining 2 -> host id

C = 1, 2, 3 octets -> net id
 4 octet -> host id

D & E no host functionality id.

IPV4 -> Internet protocol version 4
 is the 4th version of IP (Internet Protocol)
 which is used on network.

Protocol -> NetU.
 switching & type -> NetU.

10M 107 Network Models OSI

Network model is a system that explains how data is transmitted from one device to another through different layers in a network through different layers.

OSI reference model

Open system interconnection reference model
 OSI is a 7 layer reference model developed by ISO that explains how data is transmitted from one device to another in a network using different layers. It is theoretical model.

OSI reference model	Layer 1	Layer 2	Layer 3	Layer 4	Layer 5	Layer 6	Layer 7
Physical layer							
Data link layer							
Network layer							
Transport layer							
Session layer							
Presentation layer							
Application layer							

Network model is a system that divides network into layers to explain how data is transmitted from one device to another device in a network through different layers.

⑦ Application layer. ① top most layer of OSI model

⑧ App - It is a layer closest to the user the top layer.

④ In this layer the user interacts with the data through apps like browsers, email or game.

③ It is responsible for providing service to the user.

ex: browser, email

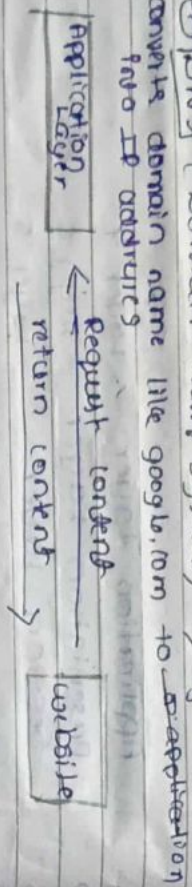
Application layer includes protocols such as

① HTTP - Hyper text transfer protocol → for web browsing

② FTP - File transfer protocol → for file transfer

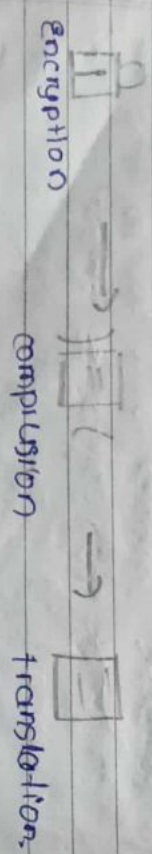
③ SMTP → Simple Mail transfer protocol → POP3 → Post office protocol for mail sending

④ DNS (Domain name system) → give name convert domain name like google.com to IP address



⑥ Presentation layer

⑥ Presentation layer take care of syntax and semantics of information exchange b/w sender and receiver



⑦ The presentation layer is responsible for converting data into a format that application layer can understand

3 main tasks

① Data Translation

② Data Encryption & Decryption

• security data before transmission

• Protects data from hackers. As they try to decrypt it, they get an error message.

③ Data Compression

• Reduce size of data, helps in faster transmission (increasing speed)

④ Data Translation

• Converts data from sender format to receiver format. Converts data from one format to another

• Data compatibility. joined to other

⑤ Session Layer

⑤ Session layer establishes, manages (maintain) synchronize and ends the communication b/w sender & receiver

⑤ logical control

• manage session b/w application, provide some dialog control. widely used

⑥ Simplex → one way communication

Keyboard → computer

⑦ Half Duplex :- Two way communication but not at same time

⑧ Full Duplex → Two way communication at same time

⑨ Connection establish - create connection b/w 2 devices

⑩ manage → keeps session alive during data transfer

⑪ end → after communication is over session ends

Segment - created by transport layer; contains data, TCP/UDP header, sequence number

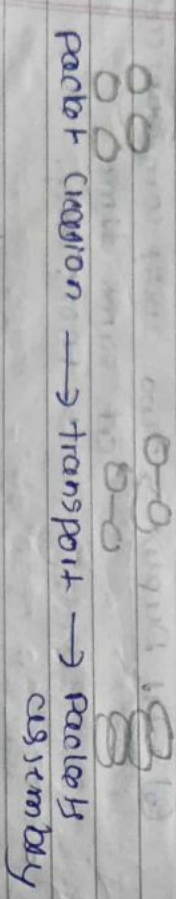
④ Transport layer (Delivery order)
 • responsible for ^{send to and delivery} packet to process delivery
 • make sure data is delivered correctly and in order, without any error

Main function

- ① Segmentation: Dividing large data into smaller parts called segments
 • makes data easy to send over network
- ② Transport (data transfer) of segments
 • transfer segments from source to destination, uses TCP & UDP protocol.
- ③ Reassembly of segments
 • reassemble / collect all segments at receiver side
 • combining them in correct order
 • recreates original message

Segmentation → Transport → Reassembly

③ Network layer, (route data)
 Network layer chooses best path for packets and sends packets from source to destination inside route.



converts segments into packets.

Packet - created by network layer, packet contains → segment + IP header
 packet = destination IP, source IP

Function

① Packet creation → each segment is divided as small as possible units called as packets / segments.
 • Each packet contains source & destination IP address

② Routing / transport of packets
 • Finds best path to transfer data
 • sends packets from source to destination

③ Packet reassembly, forwarding packets
 • At destination / receiver side, packets are reassembled, collected again to original message
 • It reassembled in a correct order
 • It routes for forwarding packets

② Data link layer
 transfer data from one device to the other device without errors
 • hop to hop, (node to node transfer)
 • This layer manages MAC (Media Access Control) (use mac address for communication)
 • It is divided into 2 sublayer
 → LLC (Logical Link Control) (frames)
 → MAC (Media Access Control) (NICs)

• Packets in the data link layer called as 'frames'

Function

① Frame creation
 • convert packets into frames
 packets + MAC address + error control = frame

② Frame Transmission

- Transfer frames via device
- Use MAC address for delivery

③ Error and flow control

- Detect errors in frame
- Retransmit lost or damaged frame
- Control speed of sender & receiver

Frame creation → transport → error & flow control

① physical layer

- send data in form of signals (0s and 1s) from one device to another
- Over a travel using wired or wireless medium
- physical connection b/w devices. Is transmission medium
- include cables, switches
- convert bit into signals

function

① Bit synchronization (timing control)

- Make sender & receiver aware of same time
- correct reading of bits

② Bit rate control (speed control)

- control how fast data sent
- sets no of bit per second

③ Physical Topology

- show how device are connected

④ Transmission mode

Path through which data transferred

tells how data flow

- ① Simplex
- ② Half Duplex
- ③ Full Duplex

10M Q1

TCP/IP Reference Model

TCP/IP Reference Model also called Internet Model. It is a 4 layer reference model developed by DARPA that explain how data is transmitted from one device to another device in a network using different layer.

TCP/IP reference model

Application layer	Layer 4
Transport layer	Layer 3
Internet layer	Layer 2
Network layer	Layer 1

④ Application Layer

Topmost layer of TCP/IP model. It provides services directly to user.

combining functions of application, presentation, session

Layer 1 of OSI

All three layers of OSI such as application, presentation, session layer are merged and form one layer called application layer.

ex: email

at application

① HTTP, HTTPS

② FTP

③ SMTP, POP3

④ DNS

Application layer

Request control website
return control

② Transport layer:

- Transport layer is responsible for end-to-end communication / packet delivery
- ensure data is received error free, in order & without any loss
- Handly segmentation & flow control.
- Protocol used
 - ① TCP → 3 line exch
 - ② UDP → User Datagram protocol

ex: downloading a file using TCP

③ Internet layer (Some to Network layer of Computer layer)

- Internet layer chooses best path for packets and sends packets from source to destination.
- decide packet route to source to destination
- converts segments to packets
- Protocol used - IP

Function

Packet creation (Some of OSI)
Routing
Packet reassembly / forwarding packet

① Network Access Layer (Data Link layer) Physical layer

Network Access layer combined Data Link & Physical layer of OSI responsible for

- ① Framing
- ② MAC addressing
- ③ Error detection

Technology used ① Ethernet ② WiFi

ex: sending data over cable or wireless signal

Comparison

difference with - 100%

7	Application layer	4	Application layer (same)
6	Presentation layer	3	Transport layer
5	Session layer	2	Internet layer / Network layer
4	Transport layer	1	Physical layer
3	Network layer		
2	Data Link layer		
1	Physical layer		

OSI

TCP/IP

Unit Chapter 2 → Transmission Media.

Cellular telephony generation of networks

Cellular telephony generation of networks
1G, 2G, 3G, 4G, 5G

(short note 5M)

1G Cellular system

A cellular system is a wireless communication system in which geographical area is divided into cells each cell is controlled by base station (tower) that provides service to mobile cellular system has evolved through different generation 1G to 5G

1G

- 1 Introduced in the 1980s
- 2 Analog communication technology
- 3 Supported only voice call
- 4 Poor voice quality & low security

2G

- 1 Introduced in 1990s
- 2 Digital communication technology
- 3 Supported voice call, SMS, MMS
- 4 Better voice quality & security than 1G

3G

- 1 Based on LTE technology
- 2 Very speed internet access
- 3 Supports online gaming, HD video streaming, video conferencing

4G

- 1 Introduced around 2010
- 2 Digital technology (LTE)
- 3 much faster than 3G

- 3 3G
- 1 Introduced in 2000s
- 2 High speed data communication
- 3 Supports internet browsing, video calling, multimedia
- 4 Faster than 2G

Explain cellular generation.

5G

- 1 Latest generation of mobile communication network
- 2 Support for devices, smart city
- 3 Support AI application
- 4 much faster than 4G
- 5 provides ultra high-speed internet

2G Introduced around 2020

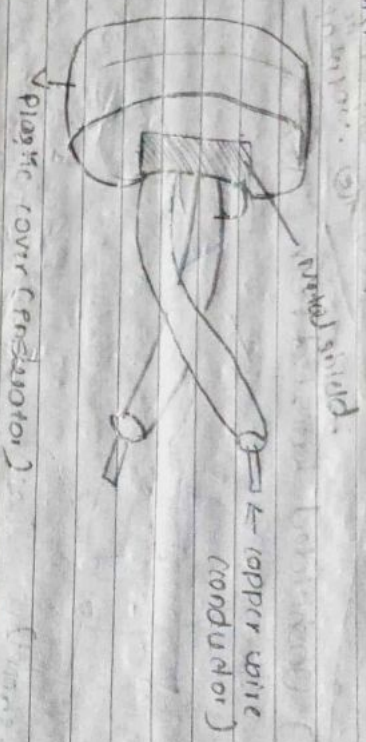
3G Advanced digital technology

Thus the cellular system has evolved from 1G to 5G from basic analog voice communication to ultra high speed digital communication

Coaxial Cable: STP is a twisted pair cable which have a extra midature shielding around the wire

② Shielded Twisted pair (STP)

- STP has a metal shield or braided mesh covering each pair of copper wiring
- It also eliminates cross-talk, which is undesired effect of one-line picking up some of the signals travelling on the other line.
- It is used in the environments with electrical interference.
- STP is used on the networks of token ring topology and medical equipments
- It is harder to handle as it is heavy & thick

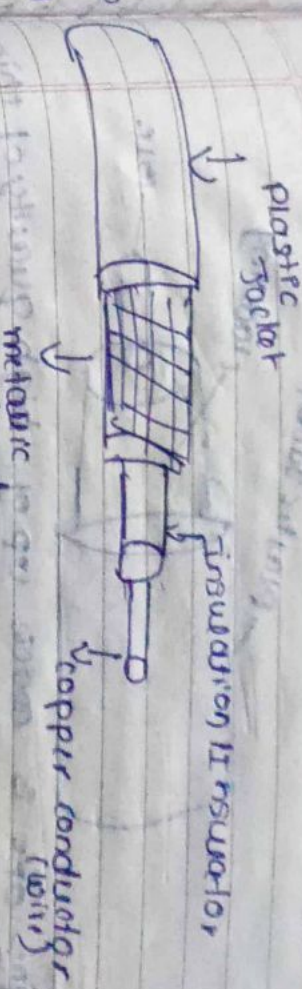


- Applications of Twisted Pair cables generally
- These cables are used within the buildings
- Used in telephone networks & data transfer
- Used in Local Area Networks (LAN)

- ① drain STP
- ② metal shield prevents lightning interference, noise
- ③ Reducing risk of data bleed out
- ④ Suitable for environments with electromagnetic noise
- ⑤ Thicker than UTP
- ⑥ More secure data transmission
- ⑦ Used in token ring topology & medical equipment

Coaxial cable consist of central inner conductor

Coaxial Cable



- Coaxial Cable - Co-axial cable is a guided transmission medium that consist of copper conductor surrounded by insulator, metallic shield and plastic jacket.
- Copper conductor wire
- Prevent at the centre of the cable
- Carry data signal

② Insulator / Insulation

- Surround copper conductor
- Prevent signal leakage

③ Metallic shield

- Surround insulator
- Protect signals from noise & cross talk

④ Plastic jacket

- The outermost layer of cable
- Protects the cable from physical damage such as fire & water

Application

- ① used in TV network
- ② used in Ethernet
- ③ used in transmission
- ④ used in high frequency

Optical Fiber Cable (OFC)



① OFC is made up of high quality of thin glass or plastic which is used to transfer data signals in the form of light up to the distance of 100 miles.

② It is not affected by any interference or noise as very low.

③ Very high speed data transmission

④ can transmit data over a long distance

⑤ lightweight & secure

Parts of OFC

① Core → central glass carries light signal

② Cladding → surround core prevents signal loss

③ Plastic jacket • outermost layer of cable • protects cable from physical damage

Applications

used in internet

• telephone

• computer network

• medical

• military

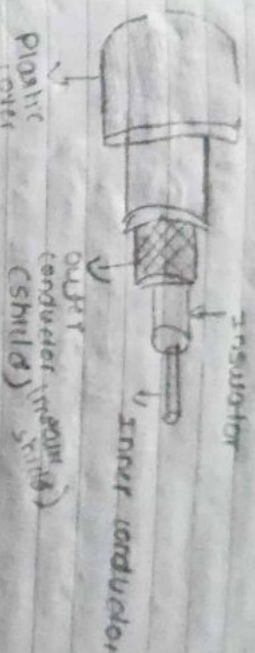
• sports

Disadvantages

• difficult to install

• expensive

Co-axial Cable



Co-axial cable consists of

• The inner copper wire which is in the center, it is enclosed in a plastic layer which provides insulation

• The outer conductor is like a shield against noise and cross-talk and this is also enclosed in a plastic shield

• The whole cable is protected by the plastic cover or jacket which is used to protect the inner layers from physical damage such as fire and water.

Applications of Co-axial cables:

• It is used for the distribution of television signals in the cable TV networks

• It is used in the analog telephone networks

• It is used in the Ethernet because of its high frequency

high frequency

• distribution

• telephone

• internet

• LAN

• Ethernet

Explain with diagram

3) Fiber optics (Optical fibre cable)



- OFC is made up of high quality of thin glass or plastic which is used to transfer data signals in the form of light upto the distance of 1000 miles.
 - ② It is not affected by any interference or noise as very low.
 - OFC carry communication signals using pulse of light generated by small lasers or light emitting diodes (LED).
 - OFC is made up of 2 or more glass strands. The center of each strand is called core which helps the light to travel.
 - The core is surrounded by a layer of glass called as cladding. That reflects the light inwards and avoid the loss of signal, so that no light escapes the glass core because of reflective cladding.
- Applications of OFC
- It is used in internet, telephone, computer network, medical, military & aerospace etc.

II) Unguided Media / Wireless Media

Unguided transmission media transmit the data signals without using physical media.

- The type of unguided media are:
 - ① Radio waves
 - ② Infrared waves

Radio waves - It is easy to generate because it has long wavelength and it can travel upto longer distance. (can pass through walls & buildings)

Infrared waves - These waves are ranging in frequency between 3 KHz to 3 GHz. (These have the high frequency and cannot be used)

③ These waves are electromagnetic waves and it can travel in all the directions.

④ Radio waves are generated by radio transmitters and received by radio receivers. Radio stations transmit radio waves using transmitters which is received by the receivers installed in our device.

⑤ Both the transmitters and receivers use an antenna to radiate and receive signals.

⑥ The applications of radio waves are: It is used in AM/FM radio, television, mobile phone.

Infrared waves - Infrared is a electromagnetic wave used for short range communication.

③ These waves frequency range is from 300 GHz to 400 THz which can be used for short range communication. (can not pass through walls)

④ These have the high frequency and can be used if there are obstacles.

⑤ They transmit data using following technology: They do not require physical cables.

④ Providing secure communication within a room
⑤ Rich interface provided to the user

• It has ~~low~~ data transmission rate

Applications of Infrared waves are

• The remote controls used in the television use infrared communication in remote control.

② It is used to transfer the data b/w a mobile ③ Air conditioner remote

④ vehicle, keyboards/mouse and speakers use infrared waves.

⑤ It is also used in automatic doors and hand-held devices.

(explain point to point protocol in detail)

Point-to-Point Protocol (PPP)

• PPP is a datalink layer protocol.

• PPP is a WAN protocol and which is commonly run over internet link.

• It is widely used in broadband communication having heavy loads & high speed.

• It is used to transmit multi protocol data between 2 directly connected (point to point) computers

• For ^{eg} considering the internet connection at ~~the~~ ^{home} user's needs to communicate b/w the computers when connected to the server of the internet service provider (ISP) and they will use PPP with the help of telephone line and modem

Frame format of PPP

Flag	8	8	16	16	8
Address	Control Protocol	Payload / data	Checksum	Flag	
Flag	Address	Control Protocol	Payload / data	Checksum	Flag

Flag: 1 byte that marks the beginning and the end of the frame

The bit pattern of the flag is 01111111

Address: 1 byte which is set to 11111111 in case of broadcast

control: 1 byte set to a constant value of 11000000. PPP doesn't provide flow control that is why this field is not used

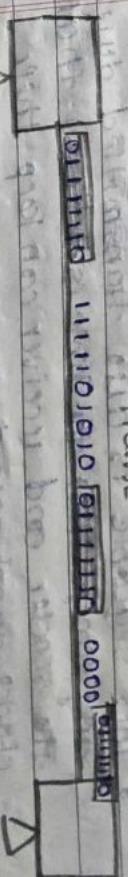
protocol: 2 or 3 bytes that define the type of data contained in the payload field

payload: This carries the data from the network layer

the maximum length of the payload field is 1500 bytes however, this may be negotiated between end points of communication.

Checksum: Error detection

Problem



Sender Receiver

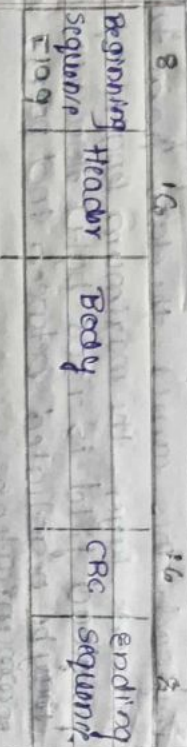
Byte stuffing or character stuffing in a data link layer is used to ensure transparent transmission. It inserts one extra byte in the data (ESC) whenever the flag sequence appears in the payload this guarantees accurate framing & data integrity

Write a short note on HDLC
Explain HDLC in detail - 10M

High-Level Data Link Control (HDLC)

- The Synchronous Data Link Control (SDLC) protocol developed by IBM is an example of a bit-oriented protocol.
- SDLC was later standardized by the ISO as the high-level data link control (HDLC) protocol.
- Bit-oriented Protocol.

HDLC - Frame Format



- The beginning sequence and ending sequence of HDLC frame is an 8-bit sequence with the bit pattern 01111110.
- This sequence is also transmitted during any time that the link is idle so that the sender and receiver can keep their clocks synchronized.
- This field indicates the beginning and ending of the frame.
- Header: This field contains address field and control field.
- Address field - If the primary station is created, then the HDLC frame address field contains 'To' address and if the secondary station is created, then the HDLC frame address field contains 'From' address.
- Control field - This control field is used for flow & error control.

- Body/Payload Field - The information field contains the user's data and other information.
- CRC - This is Error Detection Field or Frame Check Sequence (FCS) in HDLC.

Types of HDLC Frames

- There are 3 types of HDLC frames they are
- Information Frame (I-Frame) - This frame is used to transport user data and information.

Supervisory Frame (S-Frame)

This frame is used to transport control information.

Un-numbered Frame (U-Frame)

These are reserved for system management or link management.

The control field determines the type of HDLC frame.

I-Frame	1st bit is 0
S-Frame	1st two bits is 10
U-Frame	1st two bits is 11

A 1st bit 0 of the control field identifies the information frame, if the 1st 2 bits are 10 then it is identified by Supervisory frame, if the 1st 2 bits are 11 then it is identified by U-Frame or

Unnumbered Exam: Unit 4 (Medium Area content)

Channelization Protocols

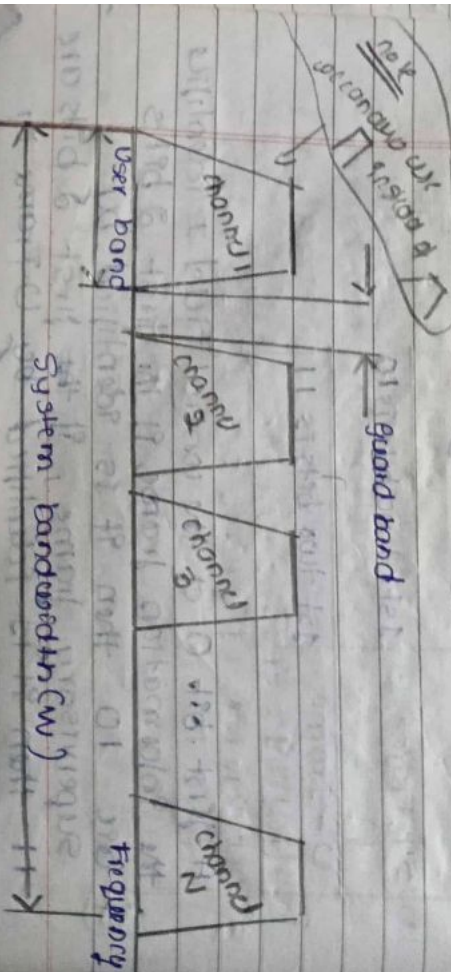
def: Channelization is a multiple access method in which the bandwidth of the channel is shared in frequency, time or code by different stations

There are 3 types of channelization they are

- ① Frequency Division Multiple Access (FDMA)
- ② Time Division Multiple Access (TDMA)
- ③ Code Division Multiple Access (CDMA)

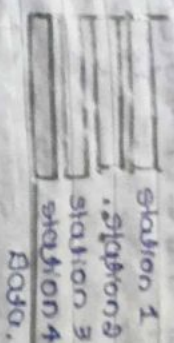
① FDMA

- In FDMA, the available bandwidth of the channel is divided into bands that are separated by guard bands
- the available bandwidth is shared by all stations.
- The FDMA is a data link layer protocol that uses FDM as the physical layer.



data

if



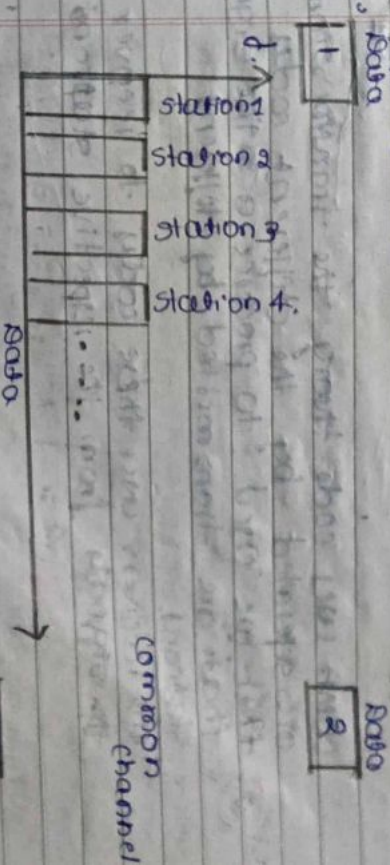
data

data

data

② TDMA

- In TDMA, the bandwidth is just one channel that is time shared between different stations.
- The entire bandwidth is just one channel, the state used by any one channel and the capacity of the channel is shared by different stations



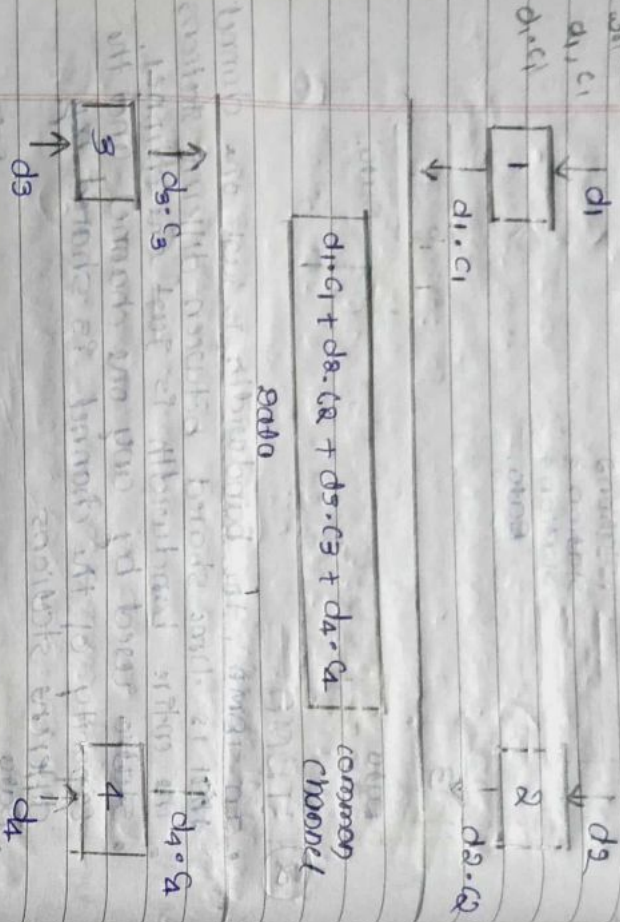
data

data

Station

③ CDMA

- In CDMA one channel carries all transmission
- Simultaneously
- CDMA differs from FDMA and TDMA because one channel occupies the entire bandwidth and no sharing of time or frequency



CDMA uses code theory - the transmission are separated by the different codes that are used to produce the signals that are transmitted by different stations. The receivers use these codes to recover the signals from the respective stations.

LAN Standards: ethernet and IEEE 802.3 LAN standard

Ethernet is a set of technologies and protocols that are used primarily for LANs (Local Area Network). It was first standardized in the 1980s by IEEE 802.3 standard (IEEE - Institute of Electrical and Electronics Engineers). IEEE 802.3 defines the physical layer & the medium access control (MAC) layer. It is a standard of the data link layer for wired ethernet networks. It is used for communication in which LANs are used. Ethernet (IEEE 802.3) Frame Format

PREPARE	S	DESTIN	SOURCE	LENGTH	DATA	CRC
F	1	6	6	2	46-1500	4
ADDRESS	1	6	6	2	46-1500	4

It indicates frame is about to begin

① PREAMBLE: Ethernet frame starts with 8 bytes preamble. This is a pattern of alternative 0's and 1's and the bit pattern is 0101010101010101.

which indicates starting of the frame and allows sender and receiver to establish bit synchronization. The preamble helps the receiver to adjust timing with the sender so that data can be received correctly.

② SFD - Start of frame delimiter. This is a 1-byte field that is always set to 10101011.

SFD indicates that the next field is the starting the frame, which is the destination address. SFD is actual beginning of ethernet frame and next field after SFD is destination address.

③ DESTINATION ADDRESS: This is a 6-byte field that contains the MAC address of the machine for which data is destined.

The frame is delivered to that destination address.

④ Source Address: This is a 6-byte field that contains the MAC address of the sender machine.

⑤ Length: This is a 2-byte field which indicates the length of the entire Ethernet frame.

This 16-bit field can hold a length value from 0 to 65535, but (length) cannot be larger than 1500 bytes because of some limitations of Ethernet.

⑥ Data: This is the core actual data is inserted, also known as payload.

Both IP header and data are placed in this field. Internet Protocol is placed here.

⑦ Minimum: Used over Ethernet. The maximum data present here is 1500 bytes and minimum data length present here is 46 bytes.

⑧ CRC - Cyclic Redundancy Check (CRC)

① CRC is a 4 byte field.

② This is used for error detection during transmission using a Cyclic Redundancy Check (CRC).

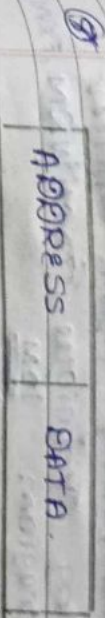
① Statistical (Asynchronous) Time Division Multiplexing

① In asynchronous TDM, the time slots are not fixed. Time slots are assigned to only the device which have data to be sent.

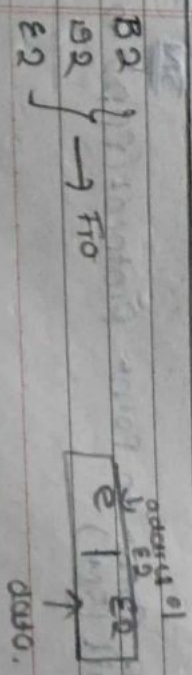
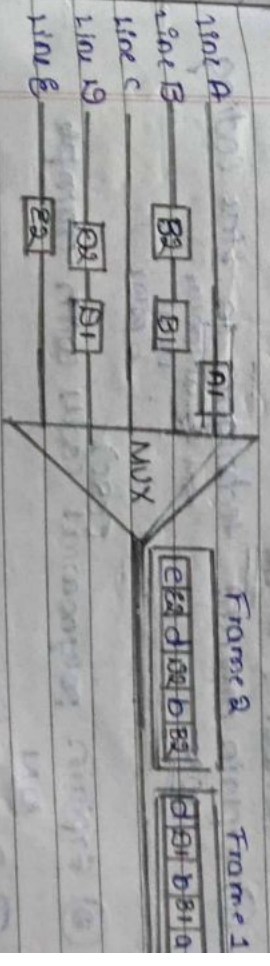
② The time slots are dynamically assigned to the device.

③ The asynchronous TDM technique fully utilizes the capacity of transmission medium.

④ Since slots are not preassigned its identity, the source of the data, an additional part is associated with the slots.



⑤ It carries the incoming data and creates a frame that contains data and without empty slots.



Important Questions

- ① What you mean by communication networks
→ SM
- ② Explain OSI reference model in detail → 10M
(with diagram)
- ③ Explain different modulation techniques → 10M
(digital modulation: BPSK, FSK, PSK, ASK, QPSK, with diagram, analog link)
- ④ What is error detection / Explain Internet checksum with an example → 5M
(take ex, explain) answer
- ⑤ What is multiplexing, explain different multiplexing techniques (FSM, WDM, FDM) with diagram 10M
- ⑥ Explain HDLC in detail 10M
(frame format and field explain)
- ⑦ What is collision? 2M
- ⑧ What is CSMA? 2M
- ⑨ Explain the methods of CSMA? 5M
- ⑩ Write a short note on Applications of Networks - 5M
- ⑪ Explain different techniques in line coding 10M
(SM → unipolar, polar, bipolar)
- ⑫ Explain polynomial code with example 10M
(CRC)
- ⑬ Define ARQ → 2M (with definition)
- ⑭ Differentiate ARQ and slotted ARQ 5M
- ⑮ Explain Point to Point Protocol (PPP) in detail 5M

- ⑯ Explain Go Back 'n' ARQ in detail. 5M
- ⑰ Define CSMA - 2M
- ⑱ Explain TCP/IP reference model 10M
- ⑲ Explain guided transmission media in detail, compare & radio
- ⑳ Explain reservation method for controlled access 5M